

Windows VM Migration

VMware to KVM — Complete Guide

Migrate Windows Server 2016-2025 and Windows 10/11 from vSphere to KVM without BSOD. Automatic VirtIO driver injection, registry fixes, and boot verification.

Zero BSOD | Auto VirtIO | AD-safe | SQL Server tested

The Windows BSOD Problem

Why Windows VMs fail after naive VMware-to-KVM migration.

Without proper driver injection, Windows VMs BSOD on first boot

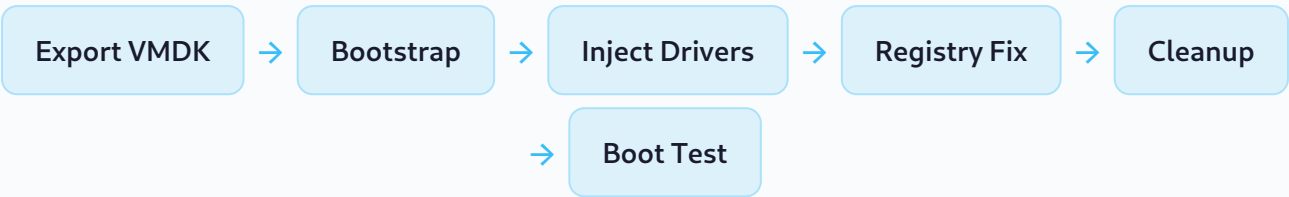
Windows kernel loads storage drivers at boot time based on registry entries. VMware uses LSI Logic or PVSCSI controllers. KVM uses VirtIO. If VirtIO drivers aren't injected before migration, Windows can't see the boot disk and blue-screens with INACCESSIBLE_BOOT_DEVICE (0x7B).

What hyper2kvm Fixes Automatically

Problem	Cause	hyper2kvm Fix
BSOD: INACCESSIBLE_BOOT_DEVICE	Missing VirtIO storage driver	Inject viostor/vioscsi from virtio-win ISO
No network after boot	VMware VMXNET3 driver, no VirtIO net	Inject NetKVM (VirtIO network driver)
High memory pressure	No balloon driver	Inject Balloon driver for memory management
Poor disk performance	Emulated IDE fallback	Configure VirtIO SCSI with proper registry entries
VMware Tools conflicts	VMware agents still running	Remove VMware Tools services and drivers
Wrong boot config (BCD)	Boot device path changed	Update BCD entries for new disk controller
Hyper-V IC conflicts	Hyper-V integration services	Remove Hyper-V IC drivers and services

Migration Pipeline

Multi-stage Windows migration with driver injection.



Supported Windows Versions

OS	Editions	Boot Mode	Status
Windows Server 2025	Standard, Datacenter	UEFI, BIOS	Fully tested
Windows Server 2022	Standard, Datacenter, Core	UEFI, BIOS	Fully tested
Windows Server 2019	Standard, Datacenter	UEFI, BIOS	Fully tested
Windows Server 2016	Standard, Datacenter	UEFI, BIOS	Fully tested
Windows 11	Pro, Enterprise	UEFI + TPM	Tested (TPM emulated)
Windows 10	Pro, Enterprise, LTSC	UEFI, BIOS	Fully tested

Example YAML Config

```
cmd: local
vmdk: windows-server-2022.vmdk
output_dir: ./out
out_format: qcow2
windows: true
guest_os: windows
virtio_win_iso: /usr/share/virtio-win/virtio-win.iso
win_stage: bootstrap
emit_domain_xml: true
vm_name: win-server-2022
memory: 4096
vcpus: 4
machine: q35
```

uefi: true

Enterprise Windows Workloads

Special considerations for critical Windows applications.

SQL Server

Migration safe: SQL Server data files migrate with the disk. No reinstall needed. VirtIO SCSI provides comparable I/O performance to PVSCSI.

Licensing win: Moving from VMware to KVM may reduce SQL Server licensing costs — KVM hosts are often counted more favorably for per-core licensing.

Active Directory DCs

Approach: Migrate secondary DCs first. Verify replication. Then migrate primary. Never migrate all DCs simultaneously.

hyper2kvm handles: NIC MAC preservation optional, DNS settings intact, AD database files migrate with disk.

Exchange Server

Tested: Exchange 2016/2019 migrated successfully. Database mounts correctly after VirtIO driver injection. Recommend maintenance mode during migration window.

IIS / Web Servers

Simplest migration: No special handling needed beyond standard VirtIO injection. Sites and app pools start automatically after boot. Verify bindings post-migration.

SQL Server Licensing Savings

Scenario	VMware (per-core)	KVM (per-core)	Savings
SQL Standard (16 cores)	\$3,945 x 16 = \$63,120	Same per-core, but fewer cores exposed	20-40% with CPU pinning
SQL Enterprise (16 cores)	\$15,123 x 16 = \$241,968	KVM allows precise vCPU control	Significant with right-sizing

Boot Verification

Every Windows migration is verified before handoff.

Automatic Boot Test

hyper2kvm starts the converted VM in KVM, waits for Windows to reach login screen, verifies no BSOD. Automated — no manual checking needed.

Console Access

VNC/noVNC console available for visual verification. See exactly what the migrated VM displays on first boot. Debug any issues in real-time.

Health Checks

Post-boot validation: disk I/O test, network connectivity, service status. Reports pass/fail for each check. Auditable results.

Rollback Ready

Source VMDKs preserved until verification passes. If anything fails, original VM is untouched on VMware. Zero-risk migration.